

Christian Aganze

✉ caganze@ucsd.edu

🌐 <https://caganze.github.io>

Appointments

- 2023–2026 ■ Stanford Science Fellow, Stanford University
2026– ■ KIPAC Rubin Fellow, Stanford University

Education

- 2016 – 2023 ■ **PhD Physics**, UC San Diego, CA
Thesis title: *Galactic Archeology with Ultracool Dwarfs*
Advisor: Prof. Adam Burgasser
2012 – 2016 ■ **B.Sc. Physics**, Morehouse College, Atlanta, GA

Publications

1184 citations, h-index = 18, i10-index = 29 (Google Scholar, June 2026)

- 1 **Aganze, C.**, Chandra, V., Wechsler, R. H., Li, T. S., Koposov, S. E., Beraldo Silva, L., Carrillo, A., Riley, A. H., Valluri, M., Gnedin, O. Y., Heiger, M., Rockosi, C., Carlberg, R., Byström, A., Kizhuprakkat, N., Lambert, M., Kim, B., Medina, G., Allende Prieto, C., . . . Zou, H. (2025, April). *The Cocytos Stream: A Disrupted Globular Cluster from our Last Major Merger?* *arXiv e-prints*, arXiv:2504.11687, arXiv:2504.11687.
<https://doi.org/10.48550/arXiv.2504.11687>
- 2 **Aganze, C.**, Pearson, S., Starkenburg, T., Contardo, G., Johnston, K. V., Tavangar, K., Price-Whelan, A. M., & Burgasser, A. J. (2024, February). *Prospects for Detecting Gaps in Globular Cluster Stellar Streams in External Galaxies with the Nancy Grace Roman Space Telescope*. *ApJ*, Vol. 9622, 151, 151. <https://doi.org/10.3847/1538-4357/ad159c>
- 3 **Aganze, C.**, Burgasser, A. J., Malkan, M., Theissen, C. A., Tejada Arevalo, R. A., Hsu, C.-C., Bardalez Gagliuffi, D. C., Ryan, R. E., & Holwerda, B. (2022a, July). *Beyond the Local Volume. II. Population Scaleheights and Ages of Ultracool Dwarfs in Deep HST/WFC3 Parallel Fields*. *ApJ*, Vol. 9341, 73, 73. <https://doi.org/10.3847/1538-4357/ac7053>
- 4 **Aganze, C.**, Burgasser, A. J., Malkan, M., Theissen, C. A., Tejada Arevalo, R. A., Hsu, C.-C., Bardalez Gagliuffi, D. C., Ryan, R. E., & Holwerda, B. (2022b, January). *Beyond the Local Volume. I. Surface Densities of Ultracool Dwarfs in Deep HST/WFC3 Parallel Fields*. *ApJ*, Vol. 9242, 114, 114. <https://doi.org/10.3847/1538-4357/ac35ea>
- 5 **Aganze, C.**, Burgasser, A. J., Faherty, J. K., Choban, C., Escala, I., Lopez, M. A., Jin, Y., Tamiya, T., Tallis, M., & Rockward, W. (2016, February). *Characterization of the Very-low-mass Secondary in the GJ 660.1AB System*. *AJ*, Vol. 1512, 46, 46.
<https://doi.org/10.3847/0004-6256/151/2/46>
- 6 Honaker, E. J., Gizis, J. E., **Aganze, C.**, Chaini, S., Bianco, F. B., Žerjal, M., Martín, E. L., Clarke, R. W., Southwick, A., Petrie, H., & Blask, T. (2026). Searching for Ultracool Dwarfs in Early LSST Data Products. *ApJ*, 1004(2), Article 214, 214.
<https://doi.org/10.3847/1538-4357/ae6ce9>

- 7 Draxl Giannoni, J. D., Desai, M., Burgasser, A. J., Dunning, A. C., **Aganze, C.**, McDermott, L., Theissen, C. A., & Bardalez Gagliuffi, D. C. (2026). Identifying and Characterizing Very Low-mass Spectral Blend Binaries with Machine Learning Methods. *AJ*, 171(3), Article 118, 118. <https://doi.org/10.3847/1538-3881/ae3086>
- 8 Lang, F. Z., Demory, B. O., Gomez Maqueo Chew, Y., Schmid, Y., Timmermans, M., Pozuelos, F. J., Gillon, M., Burdanov, A. Y., Rackham, B. V., Queloz, D., Stassun, K. G., Barkaoui, K., Triaud, A., de Wit, J., Zuniga-Fernandez, S., Burgasser, A. J., Ducrot, E., Scott, M. G., Sebastian, D., . . . Hooton, M. J. (2026). TOI-4616 b: a benchmark Earth-sized planet transiting a nearby M4 dwarf. *arXiv e-prints*, Article arXiv:2603.10905, arXiv:2603.10905. <https://doi.org/10.48550/arXiv.2603.10905>
- 9 Burgasser, A. J., Gonzales, E. C., Beiler, S. A., Visscher, C., Burningham, B., Mace, G. N., Faherty, J. K., Zhang, Z., Sousa-Silva, C., Lodieu, N., Metchev, S. A., Meisner, A., Cushing, M., Schneider, A. C., Suarez, G., Hsu, C.-C., Gerasimov, R., **Aganze, C.**, & Theissen, C. A. (2025). Observation of undepleted phosphine in the atmosphere of a low-temperature brown dwarf. *Science*, 390(6774), 697–701. <https://doi.org/10.1126/science.adu0401>
- 10 Burgasser, A. J., Schneider, A. C., Meisner, A. M., Caselden, D., Hsu, C.-C., Gerasimov, R., **Aganze, C.**, Softich, E., Karpoor, P., Theissen, C. A., Brooks, H., Bickle, T. P., Gagné, J., Artigau, È., Marsset, M., Rothermich, A., Faherty, J. K., Kirkpatrick, J. D., Kuchner, M. J., . . . Backyard Worlds: Planet 9 Collaboration. (2025). New Cold Subdwarf Discoveries from Backyard Worlds and a Metallicity Classification System for T Subdwarfs. *ApJ*, 982(2), Article 79, 79. <https://doi.org/10.3847/1538-4357/adb39f>
- 11 Barkaoui, K., Korth, J., Gaidos, E., Agol, E., Parviainen, H., Pozuelos, F. J., Palte, E., Narita, N., Grimm, S., Brady, M., Bean, J. L., Morello, G., Rackham, B. V., Burgasser, A. J., Van Grootel, V., Rojas-Ayala, B., Seifahrt, A., Marfil, E., Passegger, V. M., . . . Zúñiga-Fernández, S. (2025). TOI-2015 b: A sub-Neptune in strong gravitational interaction with an outer non-transiting planet. *A&A*, 695, Article A281, A281. <https://doi.org/10.1051/0004-6361/202452916>
- 12 Hsu, C.-C., Burgasser, A. J., Theissen, C. A., Birky, J. L., **Aganze, C.**, Gerasimov, R., Schmidt, S. J., Blake, C. H., Covey, K. R., Moreno-Hilario, E., Gelino, C. R., Serna, J., Brownstein, J. R., & Cunha, K. (2024). The Brown Dwarf Kinematics Project (BDKP). VI. Ultracool Dwarf Radial and Rotational Velocities from SDSS/APOGEE High-resolution Spectroscopy. *ApJS*, 274(2), Article 40, 40. <https://doi.org/10.3847/1538-4365/ad6b27>
- 13 Barkaoui, K., Schwarz, R. P., Narita, N., Mistry, P., Magliano, C., Hirano, T., Maity, M., Burgasser, A. J., Rackham, B. V., Murgas, F., Pozuelos, F. J., Stassun, K. G., Everett, M. E., Ciardi, D. R., Lamman, C., Pass, E. K., Bieryla, A., Aganze, C., Esparza-Borges, E., . . . Wohler, B. (2024). Three short-period Earth-sized planets around M dwarfs discovered by TESS: TOI-5720 b, TOI-6008 b, and TOI-6086 b. *A&A*, 687, Article A264, A264. <https://doi.org/10.1051/0004-6361/202349127>
- 14 Gillon, M., Pedersen, P. P., Rackham, B. V., Dransfield, G., Ducrot, E., Barkaoui, K., Burdanov, A. Y., Schroffenegger, U., Gómez Maqueo Chew, Y., Lederer, S. M., Alonso, R., Burgasser, A. J., Howell, S. B., Narita, N., de Wit, J., Demory, B.-O., Queloz, D., Triaud, A. H. M. J., Delrez, L., . . . Zong Lang, F. (2024). Detection of an Earth-sized exoplanet orbiting the nearby ultracool dwarf star SPECULOOS-3. *Nature Astronomy*, 8, 865–878. <https://doi.org/10.1038/s41550-024-02271-2>

- 15 Timmermans, M., Dransfield, G., Gillon, M., Triaud, A. H. M. J., Rackham, B. V., Aganze, C., Barkaoui, K., Briceño, C., Burgasser, A. J., Collins, K. A., Cointepas, M., Dévora-Pajares, M., Ducrot, E., Zúñiga-Fernández, S., Howell, S. B., Kaltenegger, L., Murray, C. A., Pass, E. K., Quinn, S. N., . . . Winn, J. N. (2024). TOI-4336 A b: A temperate sub-Neptune ripe for atmospheric characterization in a nearby triple M-dwarf system. *A&A*, 687, Article A48, A48. <https://doi.org/10.1051/0004-6361/202347981>
- 16 Marocco, F., Kirkpatrick, J. D., Schneider, A. C., Meisner, A. M., Popinchalk, M., Gelino, C. R., Faherty, J. K., Burgasser, A. J., Caselden, D., Gagné, J., **Aganze, C.**, Bardalez Gagliuffi, D. C., Casewell, S. L., Hsu, C.-C., Kiman, R., Eisenhardt, P. R. M., Kuchner, M. J., Stern, D., Gramaize, L., . . . Backyard Worlds Collaboration. (2024). Thirteen New M Dwarf + T Dwarf Pairs Identified with WISE/NEOWISE. *ApJ*, 967(2), Article 147, 147. <https://doi.org/10.3847/1538-4357/ad3f1d>
- 17 Rothermich, A., Faherty, J. K., Bardalez-Gagliuffi, D., Schneider, A. C., Kirkpatrick, J. D., Meisner, A. M., Burgasser, A. J., Kuchner, M., Allers, K., Gagné, J., Caselden, D., Calamari, E., Popinchalk, M., Suárez, G., Gerasimov, R., **Aganze, C.**, Softich, E., Hsu, C.-C., Karpoor, P., . . . Wedracki, Z. (2024). 89 New Ultracool Dwarf Comoving Companions Identified with the Backyard Worlds: Planet 9 Citizen Science Project. *AJ*, 167(6), Article 253, 253. <https://doi.org/10.3847/1538-3881/ad324e>
- 18 Holwerda, B. W., Hsu, C.-C., Hathi, N., Bisigello, L., de la Vega, A., Haro, P. A., Bagley, M., Dickinson, M., Finkelstein, S. L., Kartaltepe, J. S., Koekemoer, A. M., Papovich, C., Pirzkal, N., Cook, K., Robertson, C., Casey, C. M., **Aganze, C.**, Pérez-González, P. G., Lucas, R. A., . . . Kirkpatrick, A. (2024). Cosmic evolution early release science survey (CEERS): multiclassing galactic dwarf stars in the deep JWST/NIRCam. *MNRAS*, 529(2), 1067–1081. <https://doi.org/10.1093/mnras/stae316>
- 19 Kirkpatrick, J. D., Marocco, F., Gelino, C. R., Raghu, Y., Faherty, J. K., Bardalez Gagliuffi, D. C., Schurr, S. D., Apps, K., Schneider, A. C., Meisner, A. M., Kuchner, M. J., Caselden, D., Smart, R. L., Casewell, S. L., Raddi, R., Kesseli, A., Stevnbak Andersen, N., Antonini, E., Beaulieu, P., . . . The Backyard Worlds: Planet 9 Collaboration. (2024). The Initial Mass Function Based on the Full-sky 20 pc Census of ~3600 Stars and Brown Dwarfs. *ApJS*, 271(2), Article 55, 55. <https://doi.org/10.3847/1538-4365/ad24e2>
- 20 Dransfield, G., Timmermans, M., Triaud, A. H. M. J., Dévora-Pajares, M., **Aganze, C.**, Barkaoui, K., Burgasser, A. J., Collins, K. A., Cointepas, M., Ducrot, E., Günther, M. N., Howell, S. B., Murray, C. A., Niraula, P., Rackham, B. V., Sebastian, D., Stassun, K. G., Zúñiga-Fernández, S., Almenara, J. M., . . . Zapparata, A. (2024). A 1.55 $R_{\oplus}$ habitable-zone planet hosted by TOI-715, an M4 star near the ecliptic South Pole. *MNRAS*, 527(1), 35–52. <https://doi.org/10.1093/mnras/stad1439>
- 21 Ghachoui, M., Soubkiou, A., Wells, R. D., Rackham, B. V., Triaud, A. H. M. J., Sebastian, D., Giacalone, S., Stassun, K. G., Ciardi, D. R., Collins, K. A., Liu, A., Gómez Maqueo Chew, Y., Gillon, M., Benkhaldoun, Z., Delrez, L., Eastman, J. D., Demangeon, O., Barkaoui, K., Burdanov, A., . . . Schwarz, R. P. (2023). TESS discovery of a super-Earth orbiting the M-dwarf star TOI-1680. *A&A*, 677, Article A31, A31. <https://doi.org/10.1051/0004-6361/202347040>
- 22 Calissendorff, P., De Furio, M., Meyer, M., Albert, L., **Aganze, C.**, Ali-Dib, M., Bardalez Gagliuffi, D. C., Baron, F., Beichman, C. A., Burgasser, A. J., Cushing, M. C., Faherty, J. K., Fontanive, C., Gelino, C. R., Gizis, J. E., Greenbaum, A. Z., Kirkpatrick, J. D., Leggett, S. K., Martinache, F., . . . Vandal, T. (2023). JWST/NIRCam Discovery of the First Y+Y Brown

Dwarf Binary: WISE J033605.05-014350.4. *ApJL*, 947(2), Article L30, L30.
<https://doi.org/10.3847/2041-8213/acc86d>

- 23 Pozuelos, F. J., Timmermans, M., Rackham, B. V., Garcia, L. J., Burgasser, A. J., Kane, S. R., Günther, M. N., Stassun, K. G., Van Grootel, V., Dévora-Pajares, M., Luque, R., Edwards, B., Niraula, P., Schanche, N., Wells, R. D., Ducrot, E., Howell, S., Sebastian, D., Barkaoui, K., . . . Mancini, L. (2023). A super-Earth and a mini-Neptune near the 2:1 MMR straddling the radius valley around the nearby mid-M dwarf TOI-2096. *A&A*, 672, Article A70, A70.
<https://doi.org/10.1051/0004-6361/202245440>
- 24 Schneider, A. C., Burgasser, A. J., Bruursema, J., Munn, J. A., Vrba, F. J., Caselden, D., Kabatnik, M., Rothermich, A., Sainio, A., Bickle, T. P., Dahm, S. E., Meisner, A. M., Kirkpatrick, J. D., Suárez, G., Gagné, J., Faherty, J. K., Vos, J. M., Kuchner, M. J., Williams, S. J., . . . Backyard Worlds: Planet 9 Collaboration. (2023). Redder than Red: Discovery of an Exceptionally Red L/T Transition Dwarf. *ApJL*, 943(2), Article L16, L16.
<https://doi.org/10.3847/2041-8213/acb0cd>
- 25 Delrez, L., Murray, C. A., Pozuelos, F. J., Narita, N., Ducrot, E., Timmermans, M., Watanabe, N., Burgasser, A. J., Hirano, T., Rackham, B. V., Stassun, K. G., Van Grootel, V., Aganze, C., Cointepas, M., Howell, S., Kaltenegger, L., Niraula, P., Sebastian, D., Almenara, J. M., . . . Gillon, M. (2022). Two temperate super-Earths transiting a nearby late-type M dwarf. *A&A*, 667, Article A59, A59.
<https://doi.org/10.1051/0004-6361/202244041>
- 26 Gan, T., Soubkiou, A., Wang, S. X., Benkhaldoun, Z., Mao, S., Artigau, É., Fouqué, P., Arnold, L., Giacalone, S., Theissen, C. A., **Aganze, C.**, Burgasser, A., Collins, K. A., Shporer, A., Barkaoui, K., Ghachoui, M., Howell, S. B., Lamman, C., Demangeon, O. D. S., . . . Jenkins, J. M. (2022). TESS discovery of a sub-Neptune orbiting a mid-M dwarf TOI-2136. *MNRAS*, 514(3), 4120–4139. <https://doi.org/10.1093/mnras/stac1448>
- 27 Kiwy, F., Faherty, J. K., Meisner, A., Schneider, A. C., Kirkpatrick, J. D., Kuchner, M. J., Burgasser, A. J., Casewell, S., Kiman, R., Calamari, E., **Aganze, C.**, Hsu, C.-C., Sainio, A., Thakur, V., & Backyard Worlds: Planet 9 Collaboration. (2022). Discovery of 34 Low-mass Comoving Systems Using NOIRLab Source Catalog DR2. *AJ*, 164(1), Article 3, 3.
<https://doi.org/10.3847/1538-3881/ac68e7>
- 28 Ryan, R. E., Thorman, P., Aganze, C., Burgasser, A. J., Cohen, S. H., Hathi, N. P., Holwerda, B., Pirzkal, N., & Windhorst, R. A. (2022). A Self-consistent Model for Brown Dwarf Populations. *ApJ*, 932(2), Article 96, 96.
<https://doi.org/10.3847/1538-4357/ac6de5>
- 29 Softich, E., Schneider, A. C., Patience, J., Burgasser, A. J., Shkolnik, E., Faherty, J. K., Caselden, D., Meisner, A. M., Kirkpatrick, J. D., Kuchner, M. J., Gagné, J., Gagliuffi, D. B., Cushing, M. C., Casewell, S. L., **Aganze, C.**, Hsu, C.-C., Andersen, N. S., Kiwy, F., Thévenot, M., & Backyard Worlds: Planet 9 Collaboration. (2022). CWISE J014611.20-050850.0AB: The Widest Known Brown Dwarf Binary in the Field. *ApJL*, 926(2), Article L12, L12. <https://doi.org/10.3847/2041-8213/ac51d8>
- 30 Gagliano, A., Izzo, L., Kilpatrick, C. D., Mockler, B., Jacobson-Galán, W. V., Terreran, G., Dimitriadis, G., Zenati, Y., Auchettl, K., Drout, M. R., Narayan, G., Foley, R. J., Margutti, R., Rest, A., Jones, D. O., **Aganze, C.**, Aleo, P. D., Burgasser, A. J., Coulter, D. A., . . . Tinyanont, S. (2022). An Early-time Optical and Ultraviolet Excess in the Type-Ic SN 2020oi. *ApJ*, 924(2), Article 55, 55. <https://doi.org/10.3847/1538-4357/ac35ec>

- 31 Faherty, J. K., Gagné, J., Popinchalk, M., Vos, J. M., Burgasser, A. J., Schümann, J., Schneider, A. C., Kirkpatrick, J. D., Meisner, A. M., Kuchner, M. J., Bardalez Gagliuffi, D. C., Marocco, F., Caselden, D., Gonzales, E. C., Rothermich, A., Casewell, S. L., Debes, J. H., **Aganze, C.**, Ayala, A., . . . Backyard Worlds: Planet 9 Collaboration. (2021). A Wide Planetary Mass Companion Discovered through the Citizen Science Project Backyard Worlds: Planet 9. *ApJ*, 923(1), Article 48, 48. <https://doi.org/10.3847/1538-4357/ac2499>
- 32 Hsu, C.-C., Burgasser, A. J., Theissen, C. A., Gelino, C. R., Birky, J. L., Diamant, S. J. M., Bardalez Gagliuffi, D. C., **Aganze, C.**, Blake, C. H., & Faherty, J. K. (2021). The Brown Dwarf Kinematics Project (BDKP). V. Radial and Rotational Velocities of T Dwarfs from Keck/NIRSPEC High-resolution Spectroscopy. *ApJS*, 257(2), Article 45, 45. <https://doi.org/10.3847/1538-4365/ac1c7d>
- 33 Schneider, A. C., Meisner, A. M., Gagné, J., Faherty, J. K., Marocco, F., Burgasser, A. J., Kirkpatrick, J. D., Kuchner, M. J., Gramaize, L., Rothermich, A., Brooks, H., Vrba, F. J., Bardalez Gagliuffi, D., Caselden, D., Cushing, M. C., Gelino, C. R., Line, M. R., Casewell, S. L., Debes, J. H., . . . Backyard Worlds: Planet 9 Collaboration. (2021). Ross 19B: An Extremely Cold Companion Discovered via the Backyard Worlds: Planet 9 Citizen Science Project. *ApJ*, 921(2), Article 140, 140. <https://doi.org/10.3847/1538-4357/ac1c75>
- 34 Meisner, A. M., Schneider, A. C., Burgasser, A. J., Marocco, F., Line, M. R., Faherty, J. K., Kirkpatrick, J. D., Caselden, D., Kuchner, M. J., Gelino, C. R., Gagné, J., Theissen, C., Gerasimov, R., **Aganze, C.**, Hsu, C.-c., Wisniewski, J. P., Casewell, S. L., Bardalez Gagliuffi, D. C., Logsdon, S. E., . . . Backyard Worlds: Planet 9 Collaboration. (2021). New Candidate Extreme T Subdwarfs from the Backyard Worlds: Planet 9 Citizen Science Project. *ApJ*, 915(2), Article 120, 120. <https://doi.org/10.3847/1538-4357/ac013c>
- 35 Kirkpatrick, J. D., Gelino, C. R., Faherty, J. K., Meisner, A. M., Caselden, D., Schneider, A. C., Marocco, F., Cayago, A. J., Smart, R. L., Eisenhardt, P. R., Kuchner, M. J., Wright, E. L., Cushing, M. C., Allers, K. N., Bardalez Gagliuffi, D. C., Burgasser, A. J., Gagné, J., Logsdon, S. E., Martin, E. C., . . . Backyard Worlds: Planet 9 Collaboration. (2021). The Field Substellar Mass Function Based on the Full-sky 20 pc Census of 525 L, T, and Y Dwarfs. *ApJS*, 253(1), Article 7, 7. <https://doi.org/10.3847/1538-4365/abd107>
- 36 Meisner, A. M., Faherty, J. K., Kirkpatrick, J. D., Schneider, A. C., Caselden, D., Gagné, J., Kuchner, M. J., Burgasser, A. J., Casewell, S. L., Debes, J. H., Artigau, É., Bardalez Gagliuffi, D. C., Logsdon, S. E., Kiman, R., Allers, K., Hsu, C.-c., Wisniewski, J. P., Allen, M. B., Beaulieu, P., . . . Backyard Worlds: Planet 9 Collaboration. (2020). Spitzer Follow-up of Extremely Cold Brown Dwarfs Discovered by the Backyard Worlds: Planet 9 Citizen Science Project. *ApJ*, 899(2), Article 123, 123. <https://doi.org/10.3847/1538-4357/aba633>
- 37 Burgasser, A. J., Lopez, M. A., Mamajek, E. E., Gagné, J., Faherty, J. K., Tallis, M., Choban, C., Tamiya, T., Escala, I., & **Aganze, C.** (2016). The First Brown Dwarf/Planetary-mass Object in the 32 Orionis Group. *ApJ*, 820(1), Article 32, 32. <https://doi.org/10.3847/0004-637X/820/1/32>

Conference Proceedings, White Papers and Research Notes

- 38 Alvarado, E., Gerasimov, R., Burgasser, A. J., Brooks, H., **Aganze, C.**, & Theissen, C. A. (2024, May). *The Spectral ANalog of Dwarfs (SAND): New Model Atmospheres with Varying Chemistry for Galactic Archaeology with Ultracool Dwarfs. Research Notes of the American Astronomical Society*, Vol. 85, 134, 134. <https://doi.org/10.3847/2515-5172/ad4bd7>

- 39 Humphreys, A., Meisner, A. M., Burgasser, A. J., Hsu, C.-C., Schneider, A. C., Theissen, C. A., **Aganze, C.**, Gerasimov, R., Schapera, N., Kirkpatrick, J. D., Marocco, F., Faherty, J. K., Caselden, D., Kuchner, M. J., Cushing, M. C., The Backyard Worlds: Planet 9 Collaboration & The Backyard Worlds: Cool Neighbors Collaboration. (2023, August). *Spectroscopic Confirmation of the Nearby, Wide-separation L Dwarf Pair CWISE J061741.79+194512.8AB*. *Research Notes of the American Astronomical Society*, Vol. 78, 184, 184. <https://doi.org/10.3847/2515-5172/acf4a0>
- 40 Desai, M., Draxl Giannoni, J. D., Dunning, C., McDermott, L., **Aganze, C.**, Theissen, C. A., & Burgasser, A. J. (2023, January). *Identifying Ultracool Binary Systems using Machine Learning Methods*. *Research Notes of the American Astronomical Society*, Vol. 71, 13, 13. <https://doi.org/10.3847/2515-5172/acb54a>
- 41 Valencia, J. V., Burgasser, A. J., Hsu, C.-C., & **Aganze, C.** (2022, December). *Spectral Characterization of the Low-mass Companion μ Virgenes B*. *Research Notes of the American Astronomical Society*, Vol. 612, 270, 270. <https://doi.org/10.3847/2515-5172/acab6a>
- 42 Gong, Z., Nava Vega, A., Gauna Gutierrez, E., Mendiola Maytorena, A., Verdaguer, C., **Aganze, C.**, Danner, C., & Burgasser, A. J. (2022, April). *Applying Random Forest Classification to Ultracool Dwarf Discovery in Deep Surveys. I. Color Classification with SDSS, UKIDSS, and WISE Photometry*. *Research Notes of the American Astronomical Society*, Vol. 64, 74, 74. <https://doi.org/10.3847/2515-5172/ac6521>
- 43 Gutierrez, E. G., Maytorena, A. M., Gong, Z., Vega, A. N., Verdaguer, C., **Aganze, C.**, Danner, C., & Burgasser, A. J. (2022, April). *Applying Random Forest Classification to Ultracool Dwarf Discovery in Deep Surveys. II. Color Classification with PanSTARRS, 2MASS, UKIDSS, and WISE Photometry*. *Research Notes of the American Astronomical Society*, Vol. 64, 75, 75. <https://doi.org/10.3847/2515-5172/ac6522>
- 44 Gezari, S., Bentz, M., De, K., French, K. D., Meisner, A., Ntampaka, M., Jedicke, R., Patel, E., Perley, D., Sanderson, R., **Aganze, C.**, Andreoni, I., Bell, E. F., Berger, E., Dell'Antonio, I., Foley, R., Hsieh, H., Kasliwal, M., Kastner, J., . . . Williams, B. (2022, February). *R2-D2: Roman and Rubin – From Data to Discovery*. *arXiv e-prints*, arXiv:2202.12311, arXiv:2202.12311. <https://doi.org/10.48550/arXiv.2202.12311>

Honors & Awards

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| 2025 | ■ Joseph A. Johnson Award for Excellence (Honorable Mention, AIP & NSBP) |
| 2023 | ■ AAS Rodger Doxsey Travel Award |
| 2022 | ■ AAS Chambliss Poster Award (Honorable Mention)
■ Bouchet Honor Society (Honorable Mention) |
| 2021 | ■ Beth Brown Memorial Award (Honorable Mention AAS, NSBP)
■ LSST DP0 Delegate |
| 2020 | ■ Beth Brown Memorial Award (AAS, NSBP)
<i>Recognition by the AAS and the NSBP for the best presentation at the NSBP meeting</i> |
| 2018 | ■ LSSTC Data Science Fellowship |
| 2016 | ■ UC-HBCU Graduate Student Fellowship
<i>Fellowship that covers 2 years of graduate school funding</i> |
| 2015 | ■ FAMOUS Travel Grant (AAS) |

Honors & Awards (continued)

2014 ■ Poster Award (SACNAS)

Telescope & Observing Proposals

- 2025 ■ **PI**, 6.5-m Magellan Telescope — 4.5 nights awarded to obtain spectroscopy of the Cocytos stellar stream discovered in DESI
- 2024 ■ **Co-I**, JWST Cycle 3 GO-4668 (PI: A. Burgasser; 61.6 hours) — *Arcana of the Ancients: A Spectral Metallicity Survey of the Lowest-Mass Stars and Brown Dwarfs*
- 2021 ■ **PI**, APF/Shane Telescope — *Radial Velocity Monitoring of a New Peculiar Nearby M Dwarf*

Invited Talks & Presentations

- 2026 ■ **University of California, Santa Cruz, Astronomy Seminar**
Mapping the Structure of the Milky Way in Large-Scale Surveys: From the Faintest Stars and Stellar Streams to the Cosmological Context (*Invited Seminar*)
- **University of Notre Dame, Astrophysics Seminar**
Mapping the Structure of the Milky Way in Large-Scale Surveys: From the Faintest Stars and Stellar Streams to the Cosmological Context (*Invited Seminar*)
- 2024 ■ **Kavli Institute Assembly**, Los Angeles, CA
Selected KIPAC member scientist, Science Day mini-symposium (*Invited Participant*)
- **Solar Neighborhood Ultracool Dwarfs Workshop (Rubin LSST)**
Ultracool Dwarfs as Tracers of Galactic Structure with the Vera Rubin Observatory (*Invited Talk*)
- 2023 ■ **San Francisco State University, Physics & Astronomy Colloquium**
Galactic Archeology with Low-Mass Stars and Brown Dwarfs: Prospects with Large-Scale Surveys (*Invited Colloquium*)
- **University of Manchester, Astronomy Seminar**
Galactic Archeology with Ultracool Dwarfs and Stellar Streams (*Invited Seminar*)
- 2022 ■ **LSST Kickstarter Colloquium**
Counts, Colors, Kinematics and Ages for Ultracool Dwarfs with HST Surveys and the Vera Rubin Observatory (*Invited Talk*)
- **Stanford KIPAC Tea Talk**
Detecting Gaps in Globular Cluster Streams in M31 and Other External Galaxies with the Nancy Grace Roman Telescope
- **University of Michigan, Astronomy Seminar**
Galactic Archeology with Ultracool Dwarfs (*Invited Talk*)
- **National Society of Black Physicists (NSBP)**
Galactic Archeology with Ultracool Dwarfs (*Contributed Talk*)
- **NSBP Innovate Seminar**
Ultracool Dwarfs as Tracers for Galactic Structure and Star Formation History: Prospects with Large-Scale Surveys (*Invited Talk*)
- **Journal Club Talk, STScI**
Detecting Gaps in Globular Cluster Streams in M31 and Other External Galaxies with the Nancy Grace Roman Telescope (*Invited Talk*)

Invited Talks & Presentations (continued)

- **21st Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun (Cool Stars 21)**
Popsims: A Population Simulation Code for Ultracool Dwarfs throughout the Galaxy (*Poster*)
- **American Astronomical Society Meeting (AAS 240)**
Probing Gaps in Globular Cluster Streams in External Galaxies with the Nancy Grace Roman Telescope (*Poster*)
Chambliss Poster Award (Honorable Mention)
- **Division on Dynamical Astronomy Meeting**
Probing Gaps in Globular Cluster Streams in External Galaxies with the Nancy Grace Roman Telescope (*Poster*)
- 2021 ■ **National Society of Black Physicists (NSBP)**
Scale Heights & Ages of Brown Dwarfs in Deep Fields (*Talk*)
Beth Brown Memorial Award (Honorable Mention)
- **Big Apple Dynamics**
Streams, Dark Matter: Future Prospects for Galaxies Beyond the Milky Way (*Talk*)
- **American Astronomical Society Meeting (AAS 237)**
Studying Ultracool Dwarfs with the Nancy Grace Roman Space Telescope (*Invited Talk*)
- **American Astronomical Society Meeting (AAS 237)**
Brown Dwarfs in the Galaxy: Predictions for Future Wide-Field Observatories (*Poster*)
- **Journal Club Talk, Princeton**
- 2020 ■ **Journal Club Talk, NOIRLab**
- **National Society of Black Physicists (NSBP)**
Searching for Distant Ultracool Dwarfs in Deep HST/WFC3 Surveys (*Poster*)
Beth Brown Memorial Award
- 2019 ■ **Nancy Grace Roman Telescope Meeting**
Brown Dwarfs Beyond Gaia: A Deep Survey of Late-M, L, T Dwarfs with HST-WFC3 Parallel Fields (*Poster*)
- 2018 ■ **20th Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun (Cool Stars 20)**
Brown Dwarfs Beyond Gaia: A Deep Survey of Late-M, L, T Dwarfs with HST-WFC3 Parallel Fields (*Poster*)
- 2017 ■ **American Astronomical Society Meeting (AAS 228, 229)**
Toward a Comprehensive Sample of VLM Chemical Abundances with APOGEE (*Poster*)
FAMOUS Grant Award
- 2016 ■ **American Astronomical Society Meeting (AAS 227)**
Identifying Distant Brown Dwarfs in HST/WFC3 Parallel Fields (*Poster*)
- 2015 ■ **National Society of Black Physicists (NSBP)**
Characterization of the M Dwarf Binary System GJ 660.1AB (*Poster*)
- 2014 ■ **Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)**
Characterization of the M Dwarf Binary System GJ 660.1AB (*Poster*)
Best Poster Award

Public Engagement & Outreach

Public Talks

- 2025 ■ **Rubin First Look Public Lecture**, KIPAC, Stanford University
Unveiling the Milky Way Structure (*Invited Public Lecture*)
- 2024 ■ **Music of the Spheres**, Lick Observatory, Mount Hamilton, CA
Galactic Archeology: Using Stars to Map the Structure and Evolution of the Milky Way
(*Invited Public Lecture*)

Media Coverage

- 2025 ■ *Blockbuster New Vera Rubin Observatory Will Change Astronomy Forever* — Scientific American
- *See the First Breathtaking Images Captured by the Powerful New Telescope at the Rubin Observatory* — Smithsonian Magazine
- *Cocytos: A New Stellar Stream* — Astrobites
- *Rubin Observatory Will Detect Thousands of Elusive Brown Dwarfs, Unlocking Milky Way Mysteries* — NSF–DOE Rubin Observatory
- 2024 ■ *NASA’s Roman to Search for Signs of Dark Matter Clumps* — NASA
- *Rubin Observatory Will Reveal Dark Matter’s Ghostly Disruptions of Stellar Streams* — SLAC News
- 2023 ■ *How Nancy Grace Roman Will Help Understand Dark Matter* — Universe Today (Fraser Cain), interview

Mentoring

- 2026– ■ J. Valladon (Master’s student)
- 2024– ■ B. Martinez Ramirez (Cal Poly Pomona)
- 2022–2023 ■ M. Desai (UCSD), J. D. Draxl Giannoni (UCSD), C. Dunning (UCSD), E. G. Gutierrez, G. Eduardo Gauna, A. M. Maytorena, Z. Gong, C. Verdaguer (UCSD undergraduate summer researchers)

Service

- 2023– ■ DESI Milky Way Survey — helping define the dark matter science case
- 2024– ■ Via Survey — contributing to the dark matter science case
- 2024–2025 ■ NSF Panels for the Astronomy Division
- JWST Cycle 4 Proposal Reviewer
- Stanford/KIPAC committees & TACs
- 2022 ■ Referee for MNRAS and ApJ
- 2016–2022 ■ UCSD graduate student committees

Training and Workshops

- 2021 ■ Software Carpentries Instructor Training (Virtual)
- Summer School on Galactic Dynamics (The Flatiron Institute, NY, NY)

Training and Workshops (continued)

- 2020  Big Data and Deep Learning Workshop (Pittsburg Supercomputing Center)
- 2017  Kraft Observational Astronomy Workshop (Lick Observatory, San Jose, CA)
-  Scicoder Workshop (Vanderbilt University, Nashville, TN)

Skills

- Coding  Python, Github, Parallel Computing
- Research  NIR Spectral Analysis, Machine Learning, Archival Data Analysis, Optical & NIR Telescope Observations